Section: Excel rectangular section | Units: kip, kip-ft, in | Analysis: shape   Input SHA-256: fef4e5805d0b9d75...

1. General Information

Project / section	Excel rectangular section
Design code	ACI 318-19
Run option	Factored PMM strength
Capacity method	Radial contour intersection at constant Pu
Concrete integration	Shape
Neutral-axis increment	5.000 deg
Slenderness	Not considered

2. Material Properties

2.1 Concrete

Property	Value
Concrete strength, f'c	4.000 ksi
Maximum compression strain, eps_cu	0.003 in/in
Equivalent block stress	3.400 ksi
Equivalent block beta1	0.850

2.2 Reinforcing steel

Property	Value
Yield strength, fy	60.000 ksi
Elastic modulus, Es	29,000 ksi
Yield strain, eps_y	0.0020690 in/in
Stress model	Elastic-perfectly plastic

3. Section

Property	Value
Section type	Rectangular
Width x depth	20.000 x 30.000 in
Gross area, Ag	600.000 in2
Ix	45000.000 in4
Iy	20000.000 in4
Reference centroid	x = 0.000 in, y = 0.000 in

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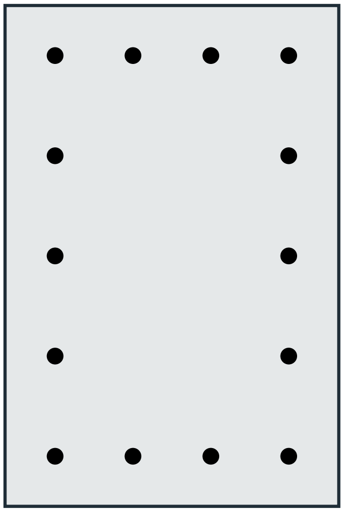
3.1 Section Figure



20.00 x 30.00 in

Figure 1: Reinforced concrete section

Section summary	Value
Dimensions	20.00 x 30.00 in
Clear cover	2.000 in
Bar centerline cover	3.000 in
Longitudinal bars	14 - #8
Steel area, Ast	11.060 in2
Reinforcement ratio	1.843%
Tie size	#4

4. Reinforcement - Bars Provided

Bar	X (in)	Y (in)	Area (in2)
B1	-7.000	12.000	0.790
B2	-2.333	12.000	0.790
B3	2.333	12.000	0.790
B4	7.000	12.000	0.790
B5	7.000	6.000	0.790
B6	7.000	0.000	0.790
B7	7.000	-6.000	0.790
B8	7.000	-12.000	0.790
B9	2.333	-12.000	0.790
B10	-2.333	-12.000	0.790
B11	-7.000	-12.000	0.790
B12	-7.000	-6.000	0.790
B13	-7.000	0.000	0.790
B14	-7.000	6.000	0.790

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5. Factored Loads and Corresponding Capacity Ratios

DCR is the demand moment radius divided by radial capacity at the same Pu.

Load	Pu	Mux	Muy	Mr,cap	DCR	Status	Note
LC-1	500.00	300.00	100.00	573.65	0.551	OK	
LC-2	900.00	350.00	180.00	477.09	0.825	OK	
LC-3	-100.00	150.00	-80.00	453.91	0.375	OK	

Moments are kip-ft; axial forces are kip. Compression is positive.

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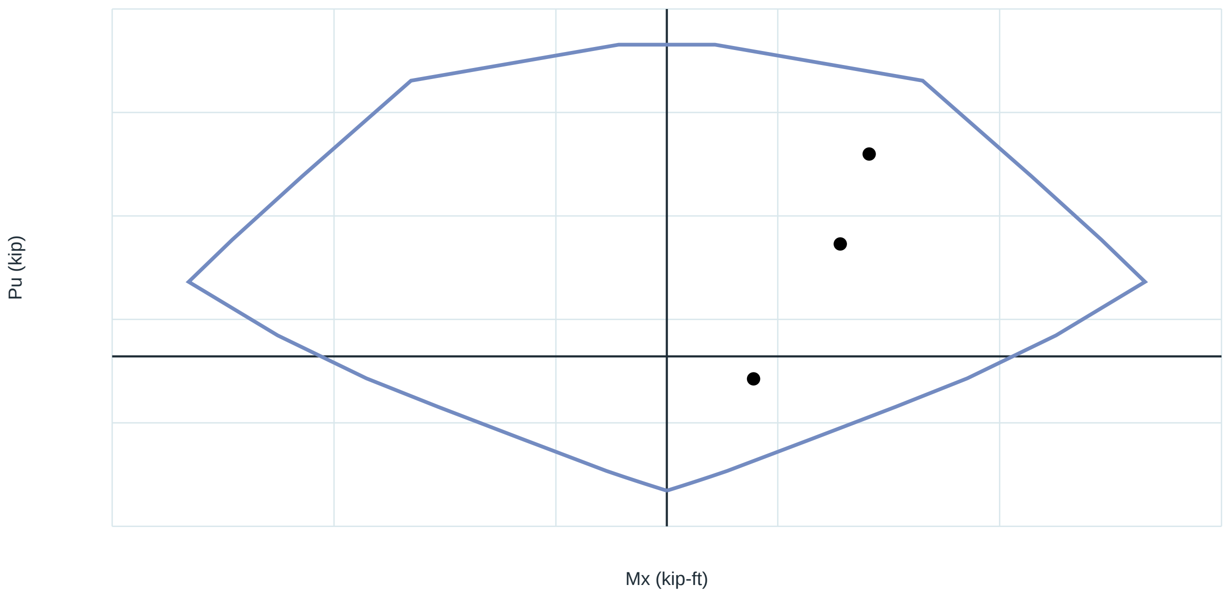
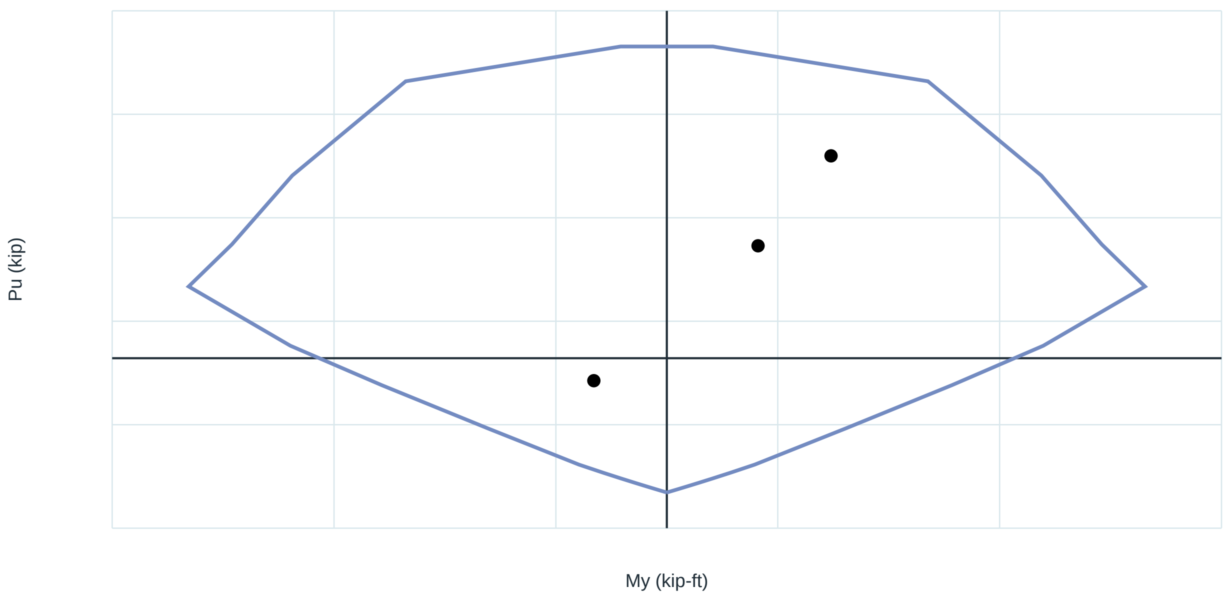
6. Diagrams

6.1 Factored P-M interaction diagrams



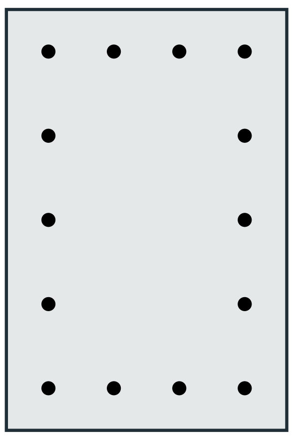


Section: Excel rectangular section | Units: kip, kip-ft, in | Analysis: shape

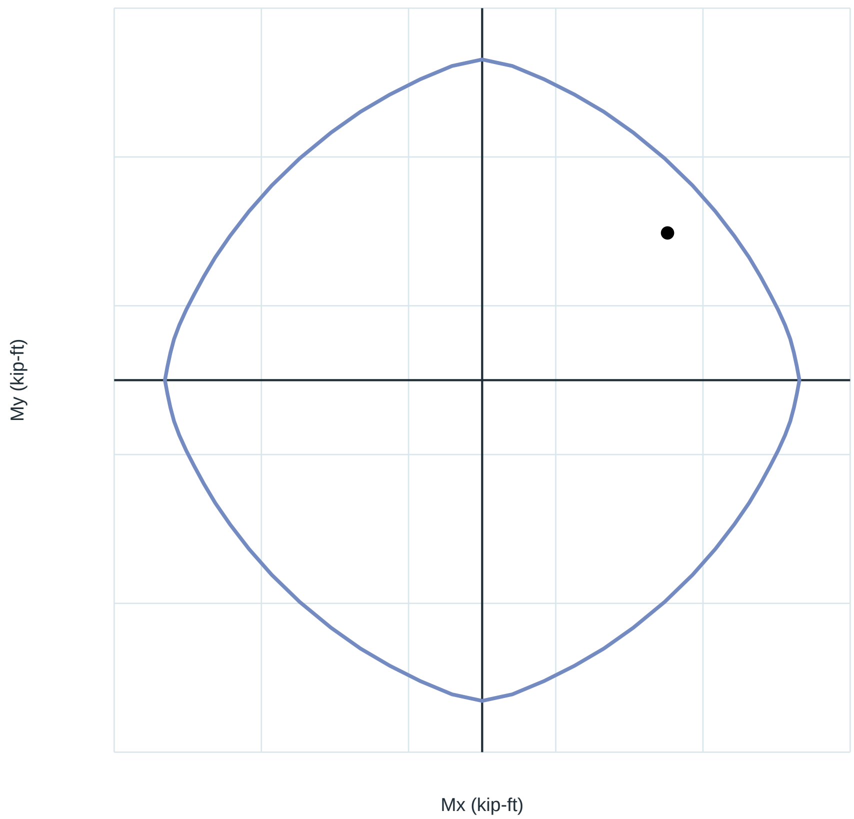
Input SHA-256: fef4e5805d0b9d75...

6.2 Constant-Pu biaxial interaction

Mx-My at Pu = 900.000 kip - LC-2



20.00 x 30.00 in



Load	Pu	Mux	Muy	Capacity radius	DCR	Status
LC-2	900.000	350.000	180.000	477.093	0.825	OK

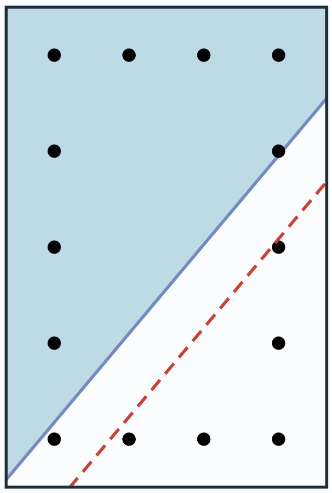
Section: Excel rectangular section | Units: kip, kip-ft, in | Analysis: shape
Input SHA-256: fef4e5805d0b9d75...

6.3 Controlling Section Strain and Stress

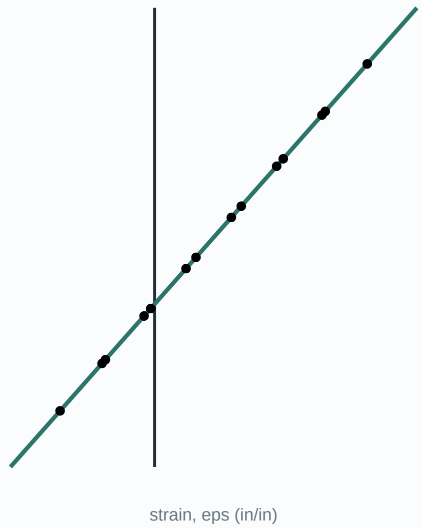
Selected load: LC-2 - compression-controlled

Closest solved orientation from the 5-degree contour; moment-direction difference = 2.087 degrees.

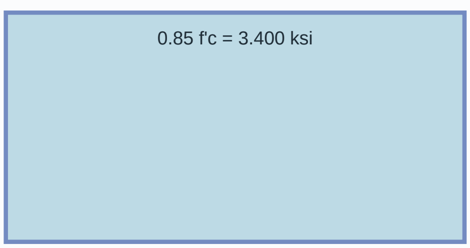
Section strain plane



Section strain distribution



ACI equivalent stress block



a = beta1 c = 18.982 in

Design idealization - not a constitutive stress-strain curve

Steel stress-strain